

23.7 Summary and Notes

23.7.1 Summary

Idea.

1. The bootstrap estimates the distribution of $\hat{\theta} - \theta$ by that of $\hat{\theta}^* - \hat{\theta}$ under an estimated mechanism: resampled residuals with the design fixed, or resampled cases (Definition 23.1.1).
2. The residual bootstrap has covariance $\hat{\sigma}_*^2(\mathbf{X}^\top \mathbf{X})^{-1}$, biased by $(n - p)/n$ (Proposition 23.1.1), and is consistent for contrasts under exchangeable errors when no case dominates (Theorem 23.1.4); it fails under heteroscedasticity.
3. The case bootstrap, linearized, has the HC0 sandwich covariance (Proposition 23.1.5). The wild bootstrap reproduces the sandwich exactly (Proposition 23.2.1) and is consistent under heteroscedasticity (Theorem 23.2.2).
4. Percentile, basic and studentized intervals are first-order correct (Corollary 23.3.2); studentizing reproduces exact pivots exactly (Proposition 23.3.3) and, by Hall's Edgeworth analysis (not proved here), gives higher-order accuracy.
5. Permutation tests are exact for any statistic under invariance (Theorem 23.4.2): in simple regression with exchangeable errors, and with nuisance regressors only for permutations that fix them (Proposition 23.4.4). Heteroscedasticity turns them into tests of independence unless the statistic is studentized.
6. The Freedman–Lane scheme is exact when an exact test exists, ignores the nuisance coefficients, and its permuted statistics are asymptotically those of an exact oracle (Proposition 23.5.1(d)); its asymptotic size is due to Anderson and Robinson (2001). Kennedy's scheme is more liberal (Proposition 23.5.2).
7. Resampling fails for extremes (Proposition 23.6.1), heavy tails, dominant cases, large p/n , dependence and, unless studentized, small samples (Section 23.6).

23.7.2 Notes and sources

The bootstrap. Efron (1979) introduced the bootstrap. Freedman (1981) set out residual and case resampling for regression and proved both consistent; Theorem 23.1.4 is a version for linear contrasts. Bickel and Freedman (1981) developed the general theory, and Wu (1986) showed that the residual bootstrap fails under heteroscedasticity. Efron and Tibshirani (1993) and Davison and Hinkley (1997) are the standard books; the latter uses the leverage-adjusted residuals adopted here. Sen and Srivastava (1990, section 5.4) describe the residual bootstrap test of a linear hypothesis.

The wild bootstrap. The idea is in Wu (1986); Liu (1988) and Mammen (1993) developed its theory, and Mammen introduced the two-point law. Davidson and Flachaire (2008) recommend Rademacher weights with restricted residuals for tests.

Intervals. Hall (1988) compared bootstrap intervals by Edgeworth expansions, and Hall (1992) is the full account, including the iterated bootstrap of Exercise (C1). Efron (1987) introduced BCa; DiCiccio and Efron (1996) review the methods. Prediction intervals for a new response, which add a resampled error to the resampled estimate in place of the normal-theory interval of

Theorem 12.3.1, are treated in Davison and Hinkley (1997, chapter 6).

Permutation tests. Fisher (1935) and Pitman (1937) introduced permutation tests. The proof of Theorem 23.4.2 follows Lehmann and Romano (2005, chapter 15); on the Monte Carlo count see Phipson and Smyth (2010). DiCiccio and Romano (2017) proved the validity of studentized permutation tests under dependence without correlation.

Partial tests. The schemes are due to Freedman and Lane (1983), ter Braak (1992), Kennedy (1995), Draper and Stoneman (1966) and Manly (1997). Anderson and Legendre (1999) compared them by simulation, Anderson and Robinson (2001) analytically, and Winkler and coauthors (2014) review them for the general linear model.

Failures. Bickel, Götze and van Zwet (1997) analyse the m -out-of- n bootstrap, Athreya (1987) the infinite-variance mean, and El Karoui and Purdom (2018) large p/n . Künsch (1989) introduced the moving-block bootstrap; Lahiri (2003) treats dependent data at length.

23.7.3 References

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